A SYNERGY OF AI OBSERVATION AND DESIGN TOOL

Leveraging Multifaceted AI Techniques for Encoding Human Behaviours and Stories in Space

CHI-LI CHENG,
Massachusetts Institute of Technology, 77 Massachusetts Avenue, Cambridge, MA, USA
Email address: chilic@mit.edu

TAKEHIKO NAGAKURA
Massachusetts Institute of Technology, 77 Massachusetts Avenue, Cambridge, MA, USA
Email address: takehiko@mit.edu

DANIEL TSAI
Massachusetts Institute of Technology, 77 Massachusetts Avenue, Cambridge, MA, USA
Email address: dtsai@mit.edu

Abstract. This paper presents an innovative AI-powered tool aimed at revolutionizing observational methods in architectural design. Its primary objective is to bridge the existing gap between designers and AI predictions, streamlining and enhancing the design process. The tool facilitates the creation of dynamic visualizations that predict human behaviours within 3D design models, adapting seamlessly to design alterations. This prototype showcases the potential for efficient AI-assisted design. The core of our system consists of an AI model that trains on data related to human behavior within environmental contexts. Our user-friendly interface empowers designers to interact dynamically with their 3D modelling tool, akin to playing an interactive chess game. Designers can populate their models with human characters, and the system, in turn, predicts the likely activities of these characters. Observational techniques are pivotal in architectural design, drawing inspiration from influential works such as those by Alexander and Whyte. They provide a comprehensive understanding of how spaces can foster human interaction and help architects, designers, and urban planners make informed decisions that enhance user-friendliness. Nevertheless, two key challenges hinder the

effective utilization of this data. Firstly, there is a lack of an intuitive interface that seamlessly integrates with existing tools. Designers often struggle to translate the information into design parameters and interpret the data effectively. Secondly, architects must adapt to evolving living environments and cultural shifts, necessitating real-time observations. However, time constraints and biases impede this process. A solution allowing designers to easily update their data is imperative. Our system comprises three integral components: a pre-trained model adaptable to specific locations, depth prediction and segmentation models for spatial comprehension, and a recognition model for user-designed structures. These features, combined with a user-friendly interface, empower designers to interact intuitively with their models, facilitating more informed and responsive design decisions.

Keywords: *AI-powered tool, Observational methods, Spatial comprehension, Human behaviours, Dynamic visualizations, Camera Based-Analysis.*

ملخص. تقدم هذه الورقة أداة مبتكرة تعمل بالذكاء الاصطناعي تهدف إلى إحداث ثورة في مناهج المشاهدة في التصميم المعماري. هدفها الأساسي هو سد الفجوة القائمة بين المصممين وتوقعات الذكاء الاصطناعي، وتبسيط وتعزيز عملية التصميم. تسهل الأداة إنشاء تصورات ديناميكية تتنبأ بالسلوكيات البشرية ضمن نماذج التصميم ثلاثية الأبعاد، وتتكيف بسلاسة مع تعديلات التصميم. يعرض هذا النموذج الأولي إمكانية التصميم الفعال بمساعدة الذكاء الاصطناعي. يتكون جوهر نظامنا من نموذج الذكاء الاصطناعي الذي يتدرب على البيانات المتعلقة بالسلوك البشري ضمن السياقات البيئية. تعمل واجهتنا سهلة الاستخدام على تمكين المصممين من التفاعل ديناميكيًا مع أداة النمذجة ثلاثية الأبعاد الخاصة بهم، تمامًا مثل لعب لعبة شطرنج تفاعلية. يستطيع المصممون ملء نماذجهم بشخصيات بشرية، ويقوم النظام بدوره بالتنبؤ بالأنشطة المحتملة لهذه الشخصيات. تعتبر تقنيات المراقبة محورية في التصميم المعماري، فهي مستوحاة من الأعمال المؤثرة مثل أعمال ألكساندر ووايت. إنها توفر فهمًا شاملاً لكيفية تعزيز المساحات للتفاعل البشري ومساعدة المهندسين المعماريين والمصممين والمخططين الحضريين على اتخاذ قرارات مستنيرة تعزز سهولة الاستخدام. ومع ذلك، هناك تحديان رئيسيان يعيقان الاستخدام الفعال لهذه البيانات. أولاً، هناك نقص في الواجهة البديهية التي تتكامل بسلاسة مع الأدوات الموجودة. غالبًا ما يواجه المصممون صعوبة في ترجمة المعلومات إلى معلمات التصميم وتفسير البيانات بشكل فعال. ثانيا، يجب على المهندسين المعماريين التكيف مع البيئات المعيشية المتطورة والتحولات الثقافية، مما يستلزم الملاحظات في الوقت الحقيقي. إلا أن ضيق الوقت والتحيزات تعيق هذه العملية. يعد الحل الذي يسمح للمصممين بتحديث بياناتهم بسهولة أمرًا ضروريًا. يتكون نظامنا من ثلاثة مكونات متكاملة: نموذج مُدرب مسبقًا وقابل للتكيف مع مواقع محددة، ونماذج التنبؤ بالعمق والتجزئة للفهم المكاني، ونموذج التعرف على الهياكل التي صممها المستخدم. تعمل هذه الميزات، جنبًا إلى جنب مع واجهة سهلة الاستخدام، على تمكين المصممين من التفاعل بشكل حدسي مع نماذجهم، مما يسهل اتخاذ قرارات تصميمية أكثر استنارة واستجابة.

الكلمات المفتاحية: *أداة تعمل بالذكاء الاصطناعى، مناهج المشاهدة، الفهم الفراغى، السلوك الإنسانى، الرؤى الديناميكية، التحليل باستخدام الكاميرات.*

1. Introduction and Overview

In the modern field of architecture, the creation of physical spaces is more than just the assembly of structures; it serves as a reflection and influence on society, culture, and individual perspectives (Borah, 2021). Contemporary architects prioritize community engagement and social consciousness in their designs, creating spaces that not only accommodate societal needs but also address important issues such as sustainability and disaster relief. The goal is to foster a sense of belonging and address modern social challenges. In this context, understanding human behavior and social dynamics in open spaces becomes crucial. Videos and photographs, now more accessible than ever through online and social media platforms, have emerged as invaluable resources for capturing these complex interactions. By analyzing this data, architects can gain insights into how people interact with their surroundings, which in turn informs designs that promote community engagement and social well-being.

In this context, we have developed a real-time analytical tool capable of predicting human behavior within the context of 3D modeling. This tool examines 3D models created within specific design software and evaluates how well the spatial design aligns with its intended purpose or program. This tool's heart is a network trained on a dataset carefully cultivated through a nuanced multi-model approach. This approach skillfully isolates key high-level features by parsing both videos and their accompanying captions using a range of pre-established models. Focusing on these essential features reduces the number of training features, which enhances the network's accuracy in predicting human behavior within different spatial contexts.

Moreover, this tool has been designed with adaptability in mind. It can be fine-tuned or retrained to align with different locales and situations. Recognizing the multifaceted nature of human behavior—shaped by cultural, environmental, and temporal determinants—this tool's adaptability ensures its relevance and effectiveness across diverse architectural design challenges.

The practical implications of this tool are far-reaching. For architects and urban planners, it can serve as a valuable assistant in creating spaces better tailored to their users' needs and behaviors. By providing insights into how people are likely to interact with a given space, designers can make more informed decisions in their design processes. This can lead to more functional and enjoyable spaces, contributing to improved quality of life for those who use them.

In addition to its applications in architectural design and urban planning, this tool holds potential for use in sociological research. Researchers can gain deeper insights into social dynamics and the factors affecting them by analyzing human behavior in various spatial contexts. This knowledge can

inform policies and initiatives to foster positive social interactions and create more inclusive and harmonious communities.

2. Introduction

In the modern field of architecture, the creation of physical spaces is more than just the assembly of structures; it serves as a reflection and influence on society, culture, and individual perspectives (Borah, 2021). Contemporary architects prioritize community engagement and social consciousness in their designs, creating spaces that not only accommodate societal needs but also address important issues such as sustainability and disaster relief. The goal is to foster a sense of belonging and address modern social challenges. In this context, understanding human behavior and social dynamics in open spaces becomes crucial. Most related research and projects are inspired by the principles outlined in Christopher Alexander's seminal work, "A Pattern Language," which investigates how architectural and urban design patterns can be used to create more harmonious and liveable spaces (Alexander, 1977). In parallel, the influential research of William H. Whyte, "The Social Life of Small Urban Spaces," holds significant value within urban planning, design, and architecture (Whyte, 1980). Whyte's work explores how people interact with and utilize public spaces, offering critical insights into the complex relationship between spatial design, human behaviour, and social dynamics.

In an era characterized by the proliferation of videos and photos, easily accessible through online and social media platforms, architects have come to recognize the invaluable potential of these resources for capturing intricate interactions and enriching the design process with valuable data. However, the seamless integration of data tools into architectural design firms faces significant challenges. Firstly, there is a pressing need for efficient techniques to organize and make sense of vast and unfamiliar datasets, thereby enhancing their practical utility. Secondly, there is a scarcity of data experts within these firms who possess the expertise to effectively manage various data sources and tools tailored to meet specific project requirements (Deutsch, 2015). While some approaches have been proposed to address these challenges, the recent emergence of AI has ushered in a revolution in the field of design, offering powerful tools to address complex problems and augment creativity (Cheng, 2017). As a result, a range of AI-related methods has started to gain prominence in addressing these needs.

MIT City Science Group studies urban areas' social, economic, and physical aspects to improve urban design practices and promote agreement among stakeholders. They introduce CityScope (2018), a data-driven platform that

simulates how urban changes affect ecosystems, allowing stakeholders to collectively assess and optimize interventions in real time. In this article, they outline the methodology of the tool, offering an innovative approach to urban design strategies.

In 2022, the SWA Group conducted the "Plaza Life Revisited" project, with the aim of examining the evolution of contemporary public spaces, revisiting them four decades after the publication of its associated book and film. The project's primary objective was to analyse shifts in how people utilize public realm spaces and identify the elements contributing to the success of these spaces. Initially, the study focused on ten plazas in Manhattan, each designed by different architects and either constructed or renovated within the past 15 years. To gain insights, the project utilized computer vision and AI algorithms to generate valuable data, including heat maps depicting dwell times, patterns of frequent and infrequent usage, as well as preliminary pedestrian counts.

In summary, the projects mentioned above have advanced data-driven design in the field of architecture, providing valuable insights. However, these approaches are currently too complex and costly for designers to readily adopt. For instance, while the visualizations are comprehensive and insightful, there exists a gap between these mediums and the design tools used by architects. Furthermore, designers may be unsure about updating or customizing the simulator and data to align with specific project requirements. This paper aims to address these challenges by developing a streamlined process to bridge this gap and make data-driven design more accessible for architects.

3. Multifaceted AI Techniques

We present an innovative tool that harnesses the power of AI to predict and analyse potential behaviors and social interactions within a designated architectural space. This cutting-edge tool seamlessly integrates with popular 3D modelling platforms like Rhino Grasshopper and Python, allowing users to interactively tag tokens representing potential human activities. As users position agents within their 3D designs, this tool provides valuable insights into the expected behaviours within the space.

Architects and designers can now engage in a dynamic, iterative design process that considers the implications of human behavior at every stage. By offering immediate feedback on how modifications to the architectural space might impact human activities and interactions, our tool empowers professionals to optimize their designs for the intended users and purposes. At the core of this tool lies a sophisticated Multifaceted AI approach, which combines a range of machine learning models and algorithms to thoroughly analyze and predict human behavior within 3D environments. This multi-model system draws from diverse data sources, including video captions,

images, and user-generated tags, to extract essential features and patterns associated with human behaviour across various spatial contexts.

The predictive capabilities of our tool are further enriched through a combination of supervised and unsupervised learning methods. Supervised learning enables the tool to refine its predictions based on labelled examples and historical data, while unsupervised learning uncovers hidden patterns and relationships within the data, revealing novel insights into potential human behaviours that may have previously gone unnoticed.

This comprehensive system (see Figure1) offers architects and designers a nuanced understanding of human behaviour within architectural spaces, enabling them to make well-informed decisions throughout the design process. By leveraging real-time feedback and insights from the tool, designers can create spaces that not only meet aesthetic and functional requirements but also facilitate positive social interactions, ultimately enhancing the overall user experience.

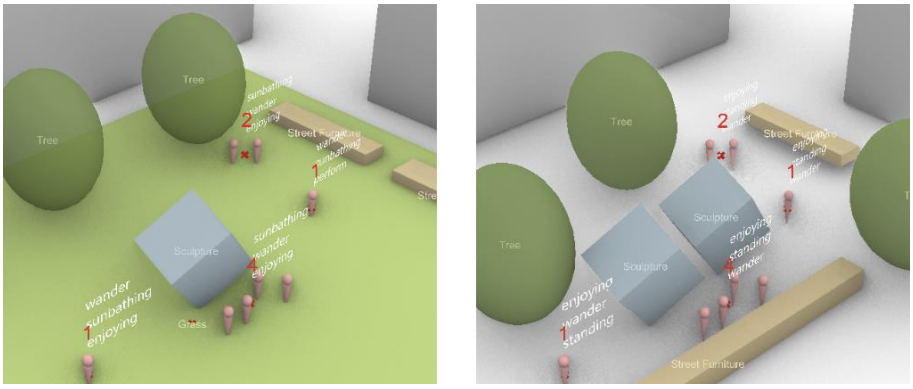

Figure 1. Dynamic Visualization of people's behavior in a design model.

Illustratively, subsequent figures depict two distinct designs eliciting varied predictions. To augment the user experience, the tool is equipped with an automatic object recognition system, bypassing the need for manual categorization of objects during estimations. This fluidity ensures that users can direct their focus on design intricacies, while simultaneously gaining insights and prompts from the estimation interface.

The underlying estimation algorithm gleans its training from genuine video footage showcasing human dynamics in open urban settings, underscoring its authenticity and applicability. Through invaluable feedback from preliminary user interactions, the tool's performance and usability have been honed.

Furthermore, the system boasts intrinsic adaptability. It can be seamlessly retrained and adjusted, providing an avenue to mold the tool's predictions to

fit varied contexts and environments. Such versatility ensures that users can align the tool's output to distinct design challenges and requirements.

4. System for Behavior Forecasting

The present system seeks to understand and predict human behavior within a given architectural space by analyzing 3D models. This system comprises two main components, as depicted in Figure 2. In the first component, our system employs a multi-model approach to extract essential and high-level information from images and captions extracted from videos. This information encompasses spatial characteristics and the behavior of individuals in the space, including their movements, interactions, and activities. By aggregating this information, the system forms a comprehensive dataset representing a wide array of architectural spaces and the behaviors of people within them. This dataset then serves as a foundation for training predictive models.

The second component of our system involves training a straightforward neural network using the dataset created in the first component. This neural network estimates and predicts people's behavior within a 3D modeling environment, as shown in Figure 3. In this 3D modeling environment, architects and designers can create virtual representations of their designs and simulate people's likely interactions with them. The neural network provides insights into how people will move through space, where they may gather, and how they will interact with their surroundings.

The system offers valuable insights into how people behave in various architectural spaces by employing both components. These insights equip architects and designers to make more informed decisions when creating their designs and to optimize spaces to meet the needs and preferences of users. Additionally, the system's adaptability allows for its application across various architectural projects, ranging from small-scale residential designs to large-scale commercial spaces. It can accommodate different types of data inputs, including images, videos, and text-based descriptions of spaces, further increasing its versatility. This adaptability establishes the system as an invaluable tool for architects and designers to create spaces that better respond to user needs.

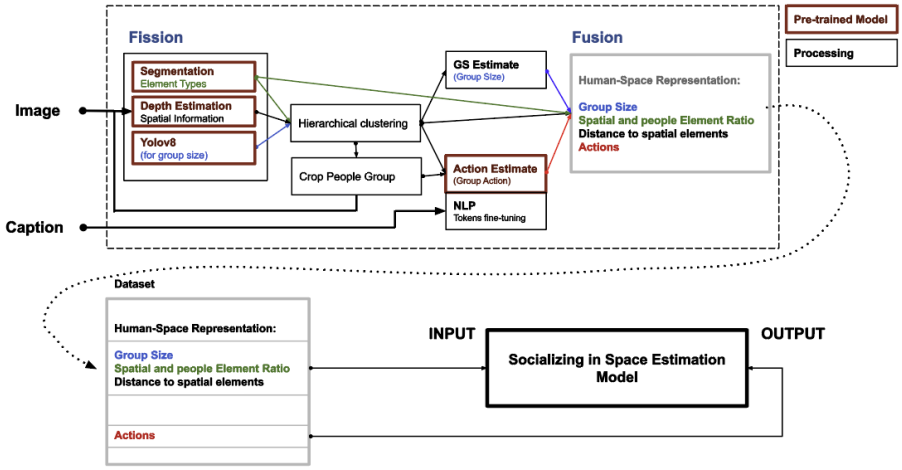

Figure 2. The system flowchart.

A multi-model technique is adopted to extract pivotal and high-tier data from images and video captions, culminating in a rich dataset.
A neural network is subsequently trained, which is geared towards estimating human behavior in a 3D modeling context (see Figure 3).

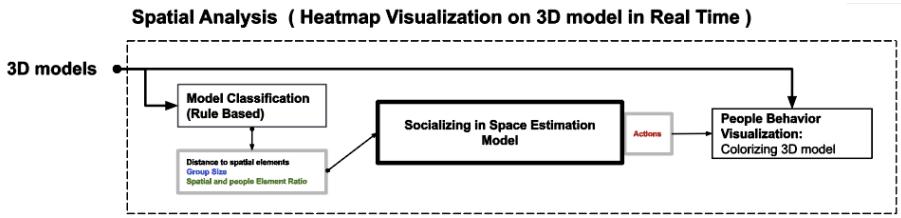

Figure 3. Spatial Analysis.

For a comprehensive analysis, the system integrates depth estimation to reproduce a 3D scene from 2D images, essential for deducing behavior in a three-dimensional setup. This is coupled with a semantic segmentation model that discerns and categorizes elements within the scene – from individuals to architectural entities like buildings and street features. This dual-model approach uncovers relationships between human behaviors and their environmental contexts. As a result, a 3D point cloud emerges, with each point assigned a distinct label. To ensure this data is cohesively organized, a hierarchical clustering technique is deployed.

Post-clustering, the system utilizes an action estimation model (based on 'Moments in Time') to deduce individual group actions (Monfort et al., 2018). Bounding boxes, marking clustered individuals, guide the system to specific image regions for precise analysis (refer to Figure 4). Additionally, the YOLO (You Only Look Once) object detection algorithm quantifies individuals within these focused image sections, determining group magnitude (Redmon et al., 2016).

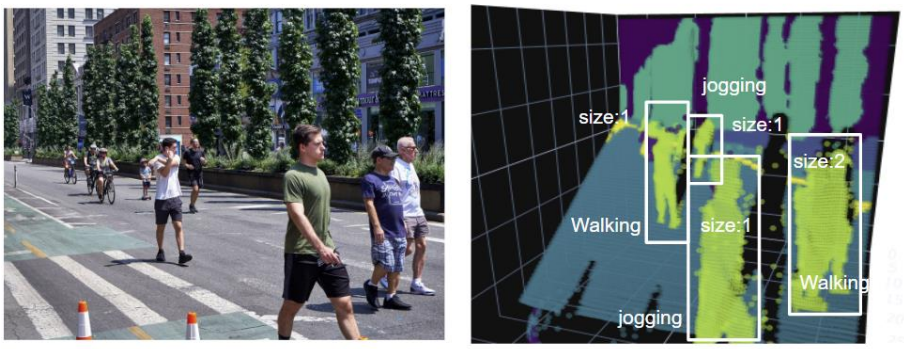

Figure 4. Bounding boxes marking clustered individuals.

We employed a combination of techniques to improve the correlation between predicted tokens and original video captions. First, we utilized T-SNE for dimensionality reduction, a technique that simplifies complex data structures without sacrificing the relationships between data points. Second, we adopted the GloVe 100D model for word embeddings, which allows us to convert words into numerical vectors that reflect their contextual meanings. Together, these methods assist us in identifying the most closely aligned token from a video's transcription, which then informs action predictions. We facilitated the transcriptions using the Google Speech-to-Text API.

After completing these procedures, our system evaluates correlations between groups and spatial elements. It does this by measuring proximities through the central points of grouped entities. As an example, if the system identifies a group engaging in a picnic near a tree, it learns that people tend to picnic close to trees. This observation contributes to the system's knowledge of inherent behaviors associated with specific spatial elements.

Our human-centric dataset for this project, depicted in Figure 5, encompasses the environmental context surrounding groups. It details their actions, group sizes, and the scene's composition in terms of element percentages. We acknowledge that behaviors are influenced by variables such as group size and environmental components. Therefore, to ensure the model's foundational strength for accurate behavior assessments, we curated a dataset of 1,000 data points. This extensive dataset captures various

behaviors and spatial contexts, making it an invaluable resource for training our predictive model.

Our system can uncover patterns and relationships between behaviors and spatial elements by analyzing the dataset (see Figure 5). This deeper understanding allows our system to make more informed predictions, aiding architects and designers in creating spaces better suited to human behavior. We intend to refine our model by incorporating additional data and implementing more advanced machine-learning techniques as we move forward. Ultimately, our goal is to provide a powerful tool that enhances the design process and helps create spaces more responsive to their users' needs and preferences.

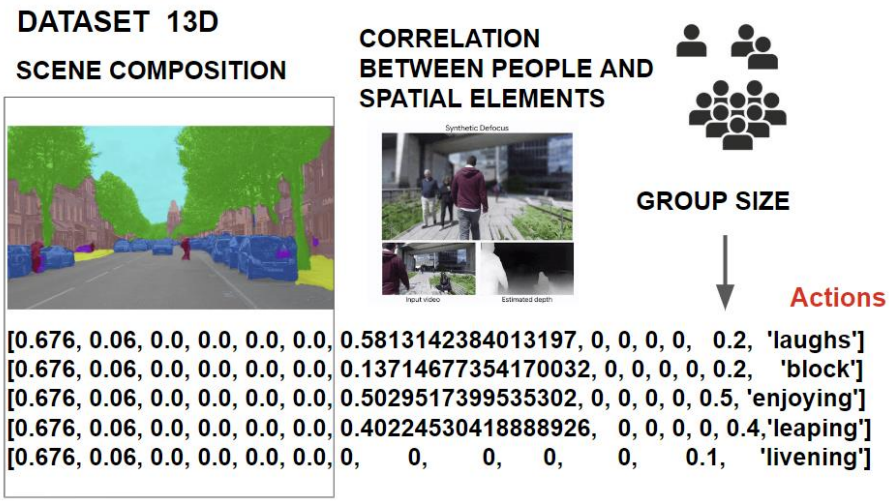

Figure 5. The dataset format

5. Implementation

We selected a video from William H. Whyte's influential documentary, "The Social Life of Small Urban Spaces" (1980). We used this documentary as a foundational reference to understand human behavior and interactions within open spaces. In this documentary, Whyte meticulously investigates the dynamics of urban public spaces, examining how people navigate and utilize areas like plazas, parks, and sidewalks. His thorough research reveals factors contributing to a space's success or failure, including its design, layout, seating arrangements, and overall ambiance. We chose this documentary for its well-organized content focusing on people's behavior in open spaces. Moreover, the captions of the documentary provide abundant information,

enabling us to employ string-processing techniques to extract relevant clips easily.

Our system gains essential knowledge about common social behaviors and patterns in urban settings. We must note that the low resolution of the documentary presented a challenge. Nonetheless, we successfully addressed this issue, showcasing our system's potential applicability to a wide variety of video sources. To break down and analyze the documentary, we used the Google speech-to-text API to convert video content into images and tokens suitable for analysis.

We then fine-tuned the model using our human-centric dataset and integrated it into Rhino Grasshopper, a powerful visual programming language equipped with a graphical algorithm editor. Grasshopper allows users to explore parametric and generative design, enabling them to create intricate shapes and patterns by interconnecting numerous components and parameters. Within this environment, users can intuitively tag tokens to represent potential activities of figures in the 3D models. As a result, our tool offers dynamic behavior predictions that update in real-time as users modify the 3D designs.

To train the neural network, we utilized the machine-learning library, Lunchbox, within Grasshopper. Lunchbox streamlines (refer to Figure 6) the training process with the provided dataset. Upon completing the training, we saved the resultant weights as a file so the Lunchbox library could access them during the estimation phase. This integration with Grasshopper enables a seamless transition from training to estimation, and users can immediately see the effects of their design modifications on human behavior predictions. This feature empowers architects and urban planners to test various design ideas and assess their impact on human interactions, helping them create spaces that encourage desired behaviors and improve overall user experiences.

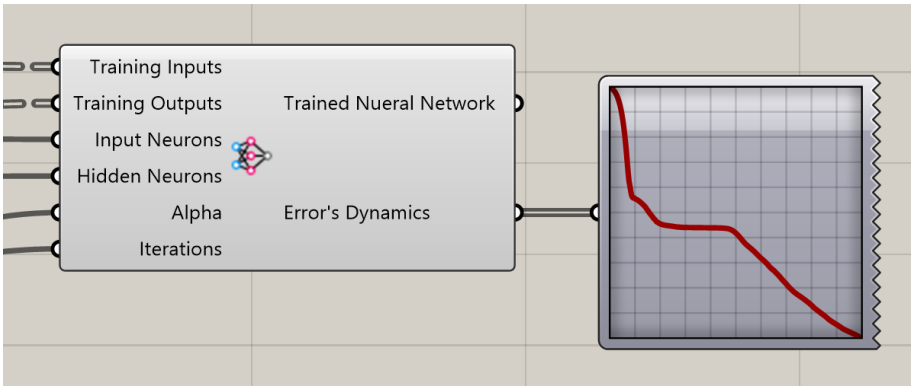

Figure 6. The Lunchbox Neural Network Component.

Ingredients of the tool development

The following provides an overview of the various libraries, tools, and packages utilized for the system, along with their specific usages and the types of input data they handle.

Library/Package/Tool:

PyTorch: This library is used for depth estimation, semantic segmentation, and activity estimation within the project. It is a popular deep learning framework that allows for flexible model creation and training.

PIL (Python Imaging Library): This library is employed for generating the dataset. It is widely used for opening, manipulating, and saving image files.

OpenCV Version 2: This library is essential for computer vision tasks, though its specific usage is not mentioned in the table.

Scikit-Learn: This machine learning library is used for fine-tuning token output along with the NLTK. It offers a wide range of tools for data mining and data analysis.

NLTK (Natural Language Toolkit): This library, used in conjunction with Scikit-Learn, is employed for fine-tuning token output. It is a leading platform for building Python programs to work with human language data.

GloVe (Global Vectors for Word Representation): While the specific usage is not mentioned in the table, GloVe is typically used for word embeddings, which can be instrumental in natural language processing tasks.

Grasshopper (in Rhino 3D): This 3D modeling platform is used to implement the network in the existing 3D modeling tool. It allows for algorithmic modeling for Rhino.

Lunchbox (Machine Learning library in Rhino 3D): This library, used in conjunction with Rhino and Grasshopper, helps implement the network in the existing 3D modeling tool.

Usages:

Pytorch: Depth Estimation, Semantic Segmentation, Activity Estimation
PIL: Generating Dataset

Scikit-Learn, and NLTK: Fine-tuning token output
Rhino, Grasshopper, and Lunchbox: Implement Network in Existing 3D Modeling Tool
Data Types of the Input:

mp4: Video files used as input for analysis.
txt: Text files, potentially used for annotations or transcriptions.
3D model in Rhino: 3D models created in Rhino that are used for spatial design and behavior prediction.

6. Conclusion

This paper has unveiled an innovative AI-powered tool with the transformative potential to reshape observational techniques in architectural design. At its core, the tool's objective is to reconcile the perceived gap between the organic design intuition of architects and the precision of AI predictions. The main advantage lies in its capacity to generate dynamic visualizations that forecast human behaviors in 3D spaces. Adjustments to designs are fluidly accommodated, allowing for immediate visualization of resultant human behavioural patterns. This tool's introduction embodies the evolution of AI-assisted design, where technology doesn't merely optimize, but also nurtures the creative essence of design, leading to spaces that resonate harmoniously with human interaction.

7. Discussion

Historically, the design process, as described by Mitchell (1990), has embraced the trial-and-error approach as intrinsic to creativity. Mirroring this philosophy, our tool grants designers the liberty to tweak designs while simultaneously observing emergent human behaviour. This real-time feedback loop promotes an integrated approach, marrying human-centric requirements with geometric considerations. Such computational aids, supported by Stiny's observation from 2001, provide a fresh vantage point, challenging designers to perceive their work differently. With this tool, designers gain deeper insights into the intricate dance between spatial designs and human behaviours. Building upon this, our methodology adopts a comprehensive strategy for deciphering human behaviour in architectural contexts. Beginning with a rich dataset representing diverse architectural spaces and associated human behaviours, we employ a gamut of data collection techniques. Post-collection, rigorous preprocessing gleans features that depict the nuanced relationship between spatial design and human dynamics. This foundational dataset serves as the bedrock for training our

predictive model, ensuring it captures the myriad ways individuals engage with architectural spaces. The distinguishing feature of our approach is the symbiotic relationship between the predictive model and the 3D modelling platform. This integration is instrumental in simulating human behaviour within projected designs, offering invaluable foresight into potential design inefficiencies or crowding hotspots.

8. Future works

While the prototype is promising, we acknowledge its static nature. It provides insights tethered to specific design alterations, potentially overlooking the fluidity of human behavior. As an advancement, we propose the integration of agent-based modeling, simulating the behavior of individual agents, each operating with unique goals. This nuanced understanding will offer more accurate projections of human interactions within architectural designs. Another avenue worth exploring is the utilization of attention-based models. Recognized for their prowess in parsing sequential data, these models could provide temporal trajectories of human behaviors, shedding light on their movement patterns and adaptations over time. Lastly, recognizing that our interface is still in its nascent stage, further enhancements are slated. Post usability testing and feedback integration, our end goal remains to furnish architects and designers with an intuitive, efficient tool to sculpt spaces that are not only aesthetically pleasing but also optimized for human inhabitation.

References

ALEXANDER, C., 1977. A Pattern Language: Towns, Buildings, Construction. 1st ed. New York: Oxford University Press.

BORAH, C., 2021. What is the Social Impact of Architecture? RTF | Rethinking The Future. Available at: https://www.re-thinkingthefuture.com/2021/10/24/a5616-what-is-the-social-impact-of-architecture/ [Accessed 15 August 2023].

CHENG, C.-Y., 2017. Interactive design process based on augmented intelligence: A framework and toolkit for designers to interact and collaborate with AI algorithms.

MITCHELL, W.J., 1990. The Logic of Architecture: Design, Computation, and Cognition. MIT Press.

MONFORT, M., et al., 2020. Moments in Time Dataset: One Million Videos for Event Understanding. IEEE Transactions on Pattern Analysis and Machine Intelligence, 42 (2), 502-508.

RANFTL, R., BOCHKOVSKIY, A., and KOLTUN, V., 2021. Vision Transformers for Dense Prediction. 2021 IEEE/CVF International Conference on Computer Vision (ICCV), 12159-12168. Available at: https://doi.org/10.1109/ICCV48922.2021.01196 [Accessed 15 August 2023].

STINY, G., 2001. New ways to look at things. Department of Architecture, Massachusetts Institute of Technology.

REDMON, J., et al., 2016. You Only Look Once: Unified, Real-Time Object Detection. IEEE Conference on Computer Vision and Pattern Recognition, 779-788.

WHYTE, W.H., 1980. The Social Life of Small Urban Spaces [film]. Directed by William H. Whyte. USA: Project for Public Spaces.

DEUTSCH, R., 2015. Data-Driven Design and Construction: 25 Strategies for Capturing, Analyzing and Applying Building Data. Wiley.

ALONSO, L., et al., 2018. Cityscope: a data-driven interactive simulation tool for urban design. Use case Volpe. International Conference on Complex Systems, 253–261. Springer, Cham.

SWA GROUP, 2023. Plaza life revisited. Available at:

https://www.swagroup.com/idea/plaza-life-revisited/ [Accessed 10 July 2023].